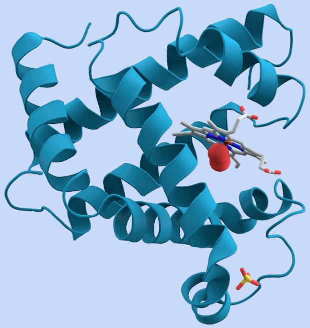
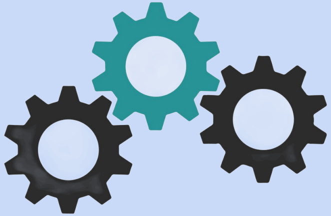


Build-A-Protein

Designing Life’s Tiny Machines



Aditi

13th Undergraduate Lecture Series

31st May, 2026

aditipathak@ncbs.res.in
https://adipat48.github.io/

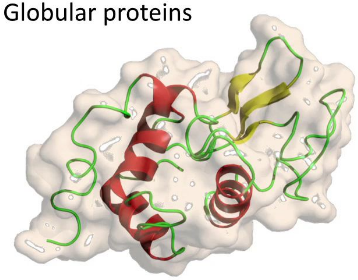
What are the different types of proteins ?

Based on Function

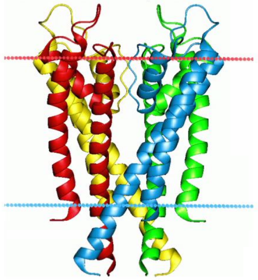
Enzymes	Structural Proteins
Transport	Regulation & Signaling
Defense-related	Contractile & Motor Proteins

Based on Structure

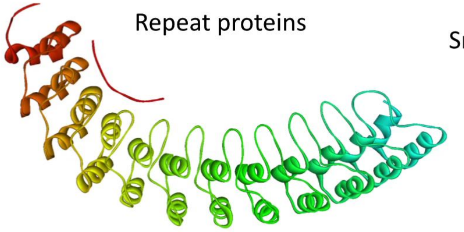
Globular proteins



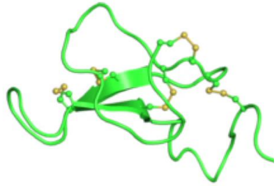
Helical bundles/channels



Repeat proteins

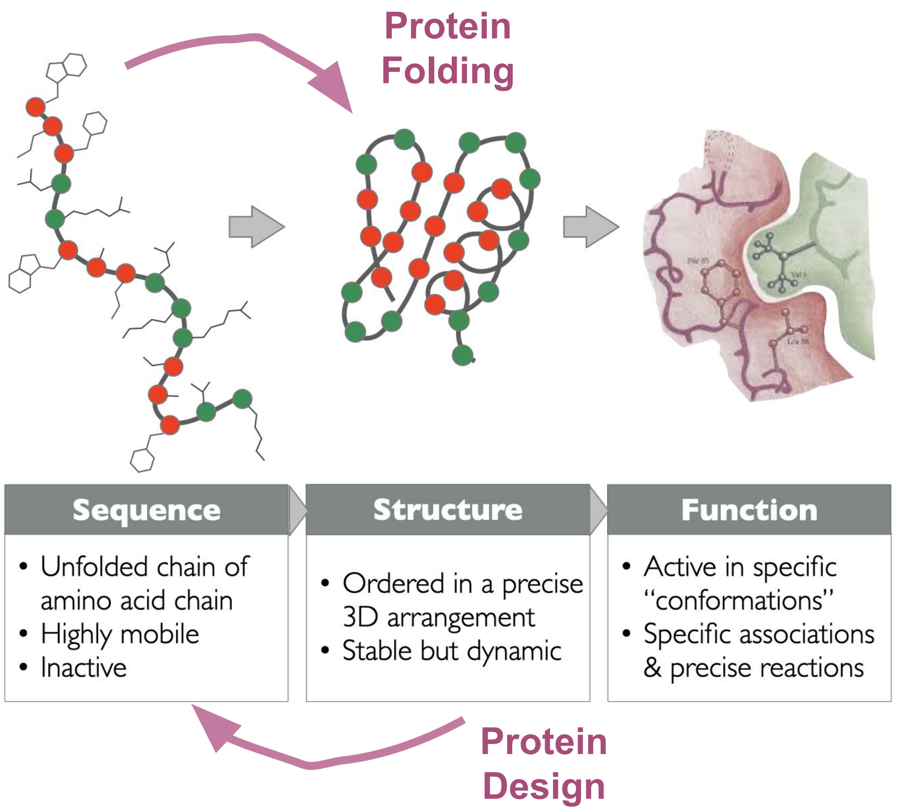


Small disulfide rich proteins



What is protein design?

Protein design seeks to create **new structures and functions** beyond those found in nature



Why do we design proteins ?

Expanding the Chemical Repertoire

New-to-Nature Catalysis
Green Chemistry
Plastic Degradation

Human Health and Therapeutics

Stability & Delivery
Neutralizing Pathogens
Vaccine Design

Enabling Experimental Biology

Fluorescence & Imaging
Thermostabilization
Cryo-EM Scaffolds

Novel Materials & Nanotechnology

Smart Materials
Enhanced Natural Fibers
Self-Assembling Cages

Intellectual Property

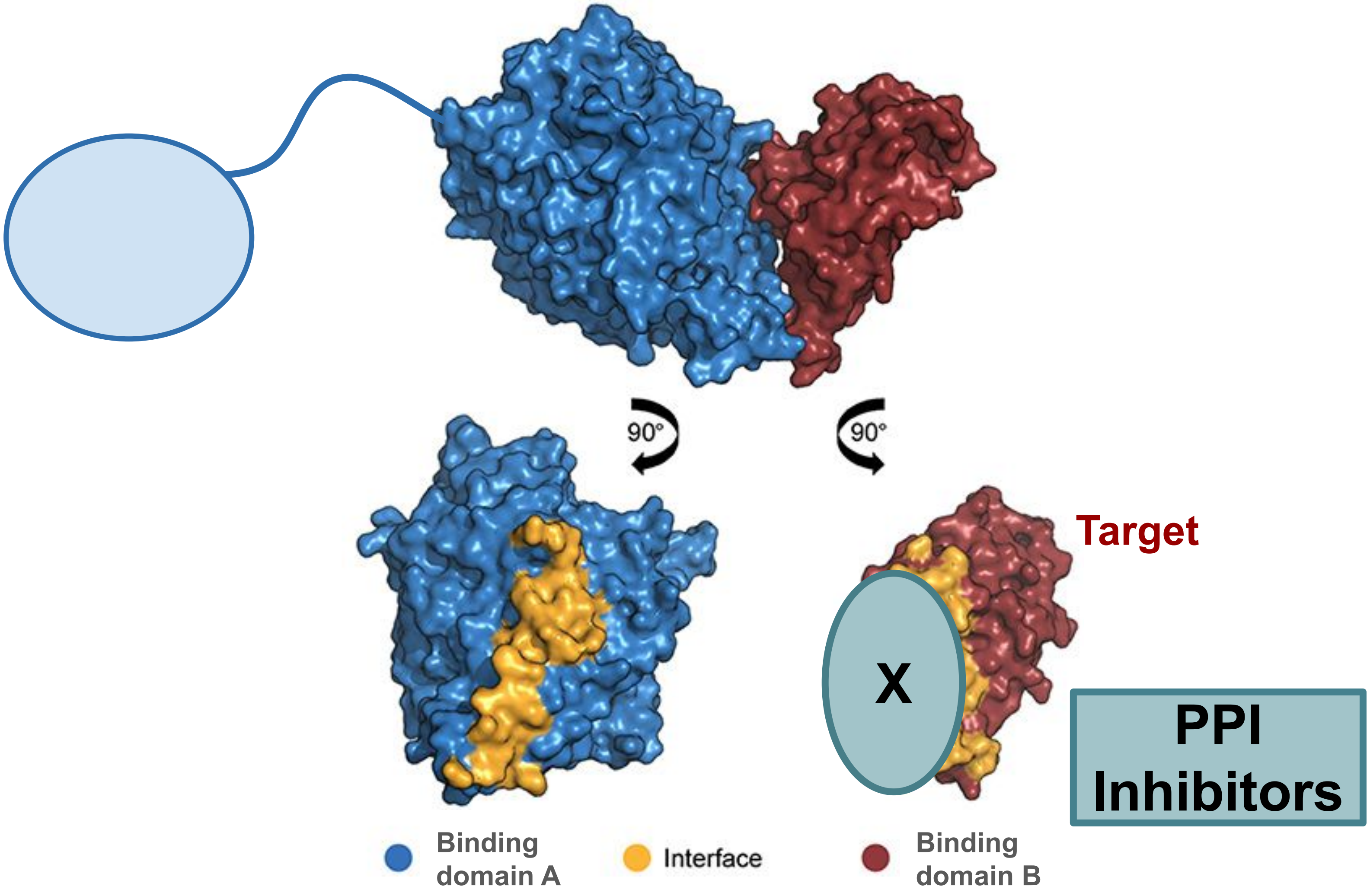
Bypassing Patents
Biobetters

Because We Can (Fundamental Research)

Biophysical Rule Book
"Dark" Sequence Space


LETS PLAY A
GAME...

Goal : De Novo Design of Proteins to Neutralize SARS-CoV-2



90°
90°
Target
X
PPI
Inhibitors
Binding
domain A
Interface
Binding
domain B

Instructions

- Each team will pull a "**Fate" Card**, which determines the specific starting budget of resources and time units your laboratory must work with.
- The goal of this game is **maximizing both your Novelty and Confidence points** while carefully avoiding bankruptcy by not running out of money and time resources.
- We will handle a **tie on the final leaderboard** by adding any remaining money and time surplus to your final score to determine the absolute winner.
- We will divide each **four-person laboratory** into roles :
  - **Project Leader** - in charge of driving decisions and consulting the team
  - **Lab Manager** - in charge of maintaining the team balance sheet ledger
  - **Experimentalist** - in charge of doing the 'experiment' (filling out worksheet, cutting, and coloring)
  - **Corresponder** - answers any questions we pose during the lecture, defends the team's choices

Instructions

- I will explain your **available options at the start of each round**, and your team must select at least one option

- We require your team to actively huddle and try to **estimate how much time and resources** you will lose versus what **point metrics you will gain** before we reveal the true costs of each technique.

- We expect all team members to be actively consulted during every major design decision, but the **Project Leader's choice** will serve as the definitive tie-breaker vote if the group splits.

Instructions

- We will offer opportunities to score **bonus confidence points** at the end of each round by asking 1 to 2 groups to stand up and explain the **pros of their choices**, followed by asking other groups to act as peer reviewers and **critique the cons** of that technique.

- We will award your team **+5 bonus points** if your peer review arguments match the hidden pros and cons on our master slide, and an impressive **+10 points** if you raise a highly valid scientific point that we haven't mentioned

- We will introduce a random **"Action Card"** in the middle of the game that your laboratory will have to deal with - for good or for worse !

Fate Cards

1

The Graduate Student

You have bare funding and a diet powered by free food at seminars, but you possess an infinite runway of absolute caffeine-fueled time

Starting Resources: 12 RU / 10 TU

2

The Startup Founder

The venture capitalists just wired the seed round, so you are drowning in cash but burning through an incredibly aggressive runway clock.

Starting Resources: 14 RU / 8 TU

3

The Big Pharma Researcher

Backed by a massive corporate empire, you have deep institutional pockets and a solid timeline, as long as you can survive the endless legal meetings.

Starting Resources: 16 RU / 9 TU

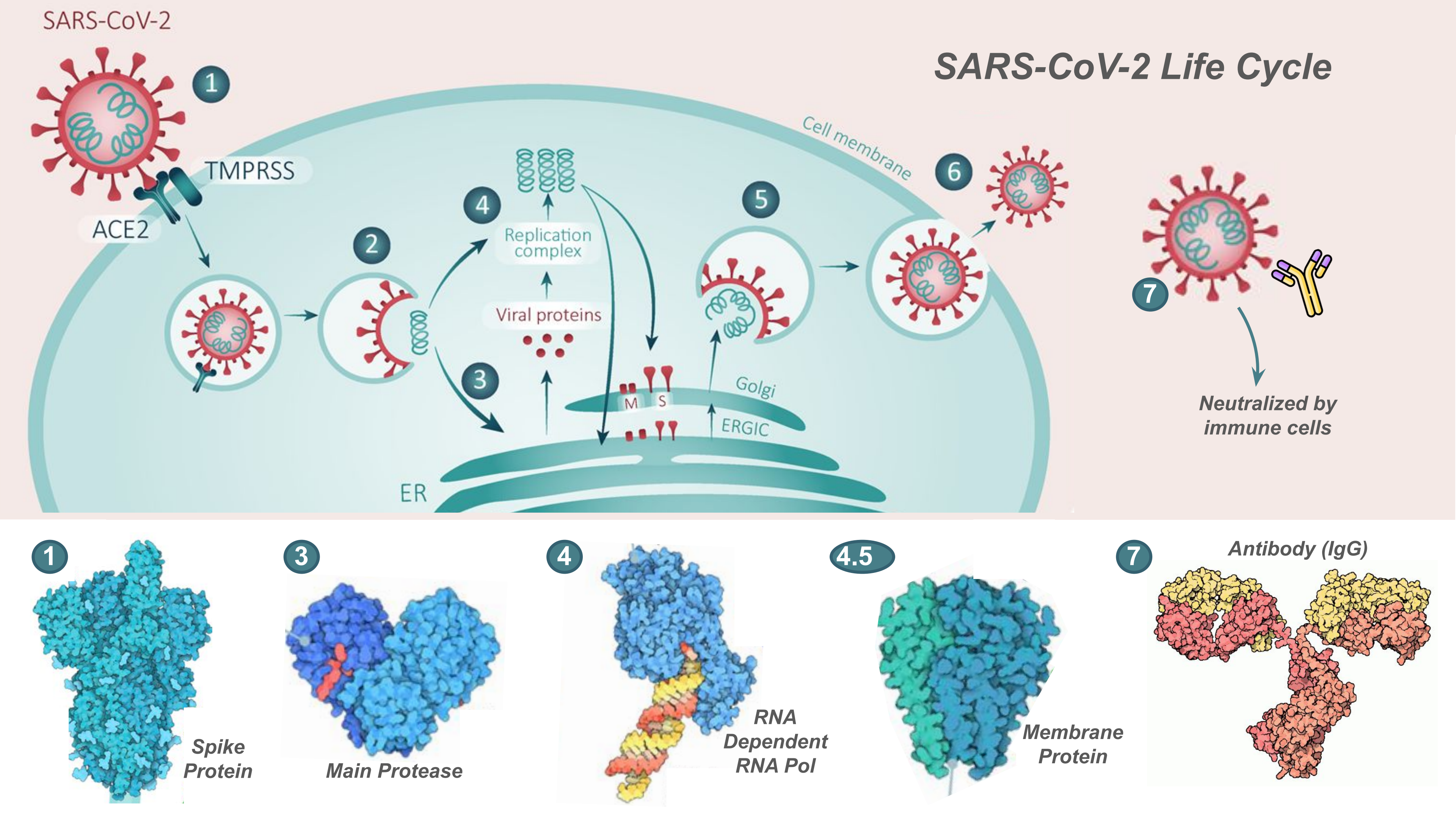
Action Cards

A

B

C

D


SARS-CoV-2 Life Cycle
SARS-CoV-2
1
TMPRSS
ACE2
2
3
4
Replication complex
Viral proteins
Cell membrane
5
6
7
Golgi
ERGIC
M
S
ER
Neutralized by immune cells
1
Spike Protein
3
Main Protease
4
RNA Dependent RNA Pol
4.5
Membrane Protein
7
Antibody (IgG)

Target Selection

Target is exposed to extracellular environment

+ 10 Confidence

Target is a viral protein

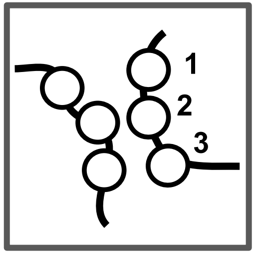
+ 10 Confidence

Target is a human protein

- 10 Confidence

Goal : De Novo Design of Proteins to Neutralize SARS-CoV-2

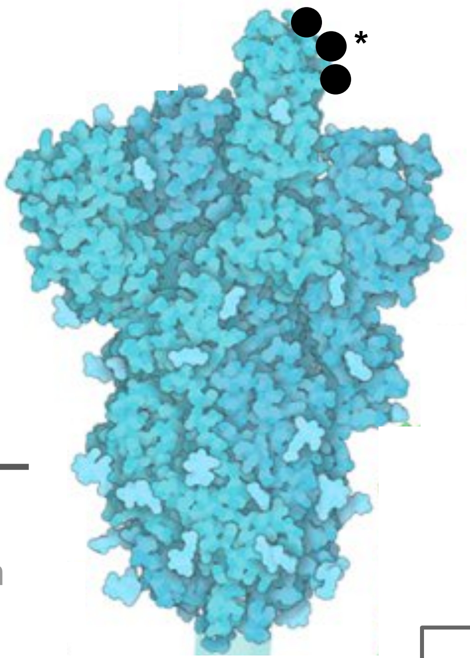
Amino Acid Type	Amino Acid	1 Letter Code
Hydrophobic: Aliphatic	Glycine	G
	Alanine	A
	Valine	V
	Leucine	L
	Isoleucine	I
	Proline	P
	Methionine	M
Hydrophobic: Aromatic	Phenylalanine	F
	Tyrosine	Y
	Tryptophan	W
Hydrophilic: Polar Uncharged	Serine	S
	Threonine	T
	Cysteine	C
	Asparagine	N
	Glutamine	Q
Hydrophilic: Acidic	Aspartic Acid	D
	Glutamic Acid	E
Hydrophilic: Basic	Arginine	R
	Histidine	H
	Lysine	K



* Binding Interface

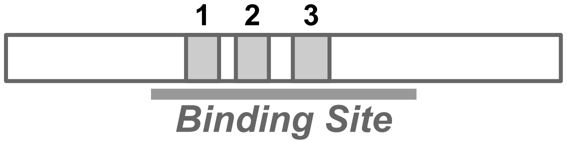
Target

Subcellular Localisation



Scaffold

Final Sequence



Scaffold Selection

1

Natural Mimic

UniProt

Protein BLAST
protein ▶ protein

RCSB PDB
PROTEIN DATA BANK

Structure Summary | Structure | Annotations | Experiment | Sequence | Genome | Ligands | Versions

Biological Assembly 1

Display Files | Download Files | Data API

1R42 | pdb_00001r42

Native Human Angiotensin Converting Enzyme-Related Carboxypeptidase (ACE2)

PDB DOI: https://doi.org/10.2210/pdb1R42/pdb

Classification: HYDROLASE
Organism(s): Homo sapiens
Expression System: Spodoptera frugiperda
Mutation(s): No

Deposited: 2003-10-07 **Released:** 2004-02-03
Deposition Author(s): Towler, P., Staker, B., Prasad, S.G., Menon, S., Ryan, D., Tang, J., Parsons, T., Fisher, M., Williams, D., Dales, N.A., Patane, M.A., Pantoliano, M.W.

Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (NAG)

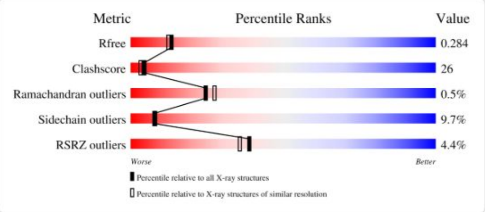
Global Symmetry: Asymmetric - C1
Global Stoichiometry: Hetero 5-mer - A2B1C1D1

Pseudo Symmetry: Asymmetric - C1
Pseudo Stoichiometry: Hetero 5-mer - A1B1C1D1E1

Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 2.20 Å
R-Value Free:
0.287 (Depositor), 0.284 (DCC)
R-Value Work:
0.235 (Depositor), 0.233 (DCC)

wwPDB Validation 3D Report | Full Report

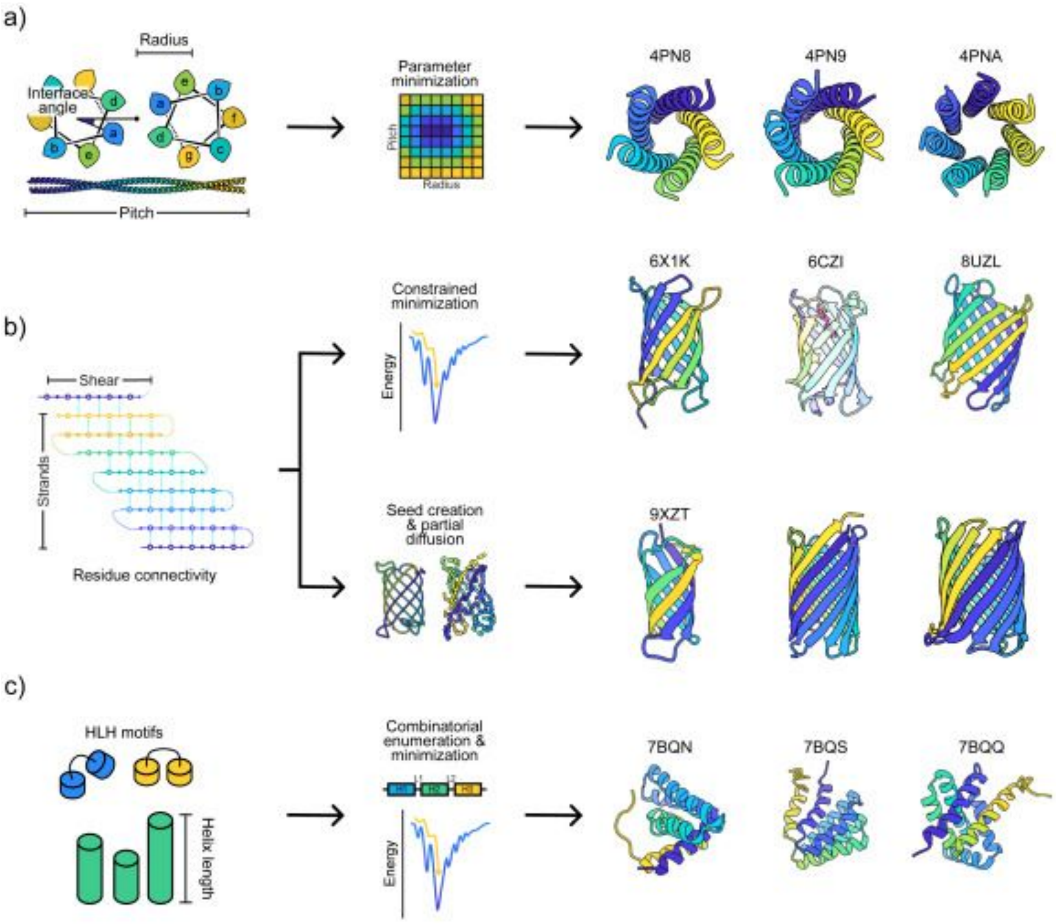


Ligand Structure Quality Assessment

Scaffold Selection

2

De novo designed Scaffolds



Scaffold Selection

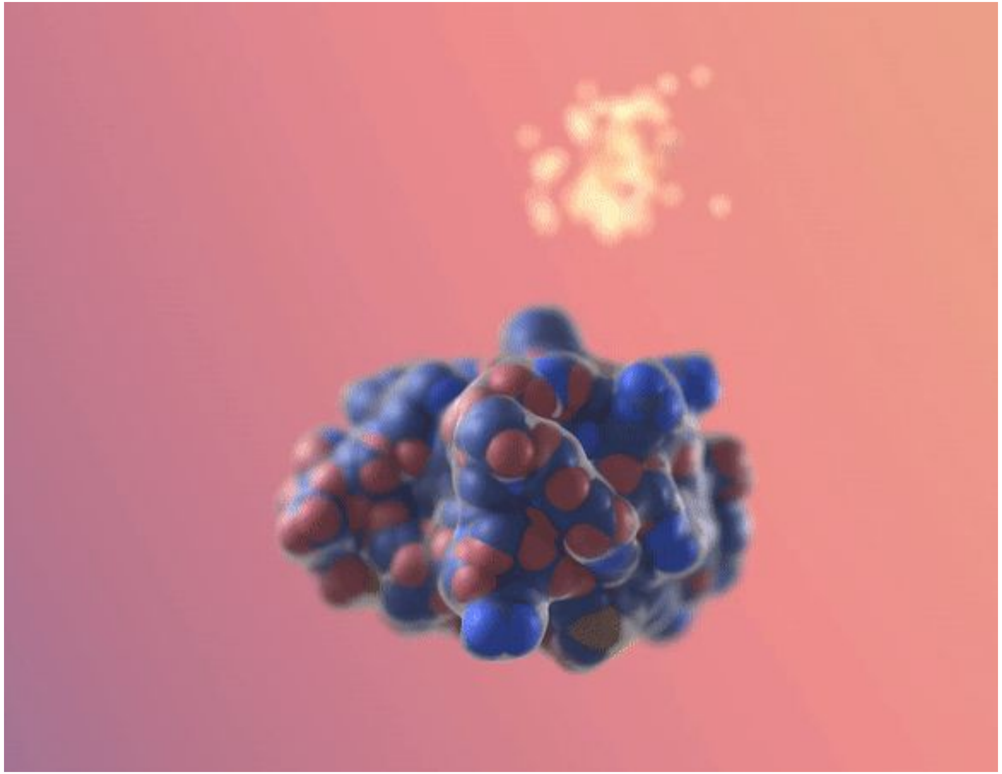
3 Generative AI

THE NOBEL PRIZE
IN CHEMISTRY 2024

David
Baker
"for computational
protein design"

Rosetta
Commons

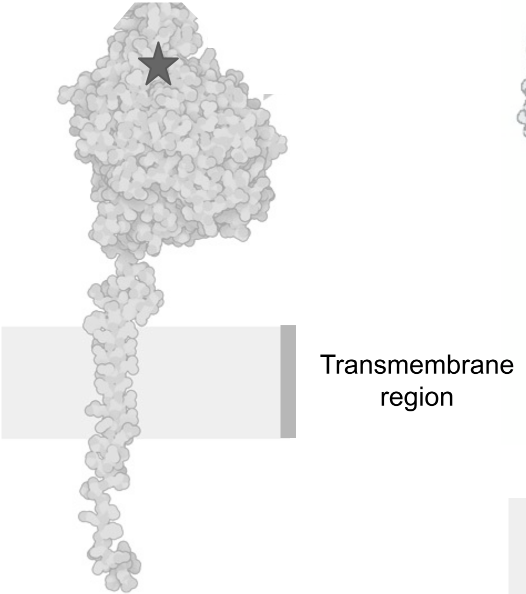
RF*diffusion*



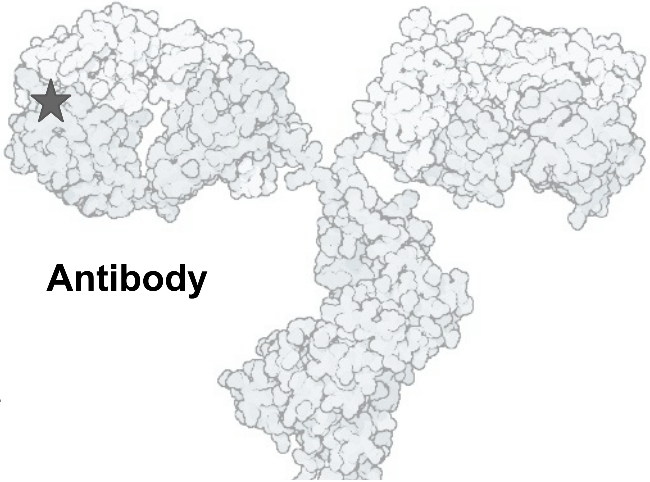
Protein Scaffolds

Binding domain

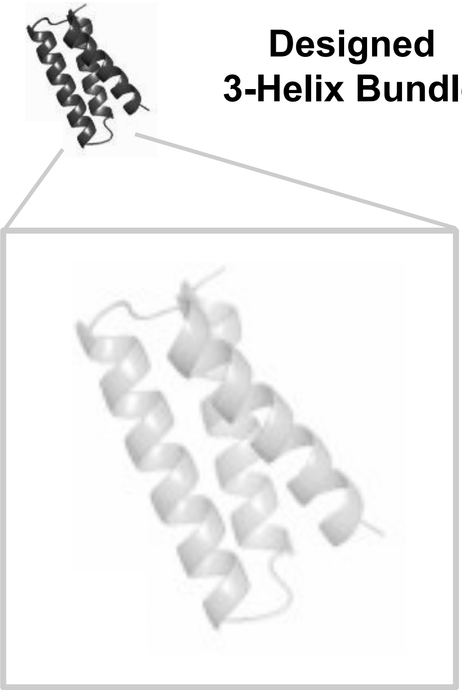
ACE2 Receptor



Antibody



Designed 3-Helix Bundle



If you chose "Generative AI", draw your own scaffold!

These structures are not to scale or completely accurate, they are only for representation purposes for this activity

Scaffold Selection

1

Natural Mimic

2

De novo designed Scaffolds

3

Generative AI

Scaffold Selection

1

Natural Mimic

Pros :
Pre-optimized complementary
Low immunogenicity

Cons :
Off-target effects
Pre-existing host cleavage sites

-1 TU | - 1 RU
+10 Novelty
+40 Confidence

Used the entire protein ?
Expression is going to be difficult

Lose -20 Confidence

Scaffold Selection

1

Natural Mimic

Pros :
Pre-optimized complementary
Low immunogenicity

Cons :
Off-target effects
Pre-existing host cleavage sites

-1 TU | - 1 RU
+10 Novelty
+40 Confidence

2

De novo designed Scaffolds

Pros :
Hyper-stability
High expression

Cons :
Structural rigidity
Binding complementarity needs to be induced

-1 TU | - 2 RU
+20 Novelty
+30 Confidence

3

Generative AI

Pros :
Non-native topologies
Precise conditioning possible

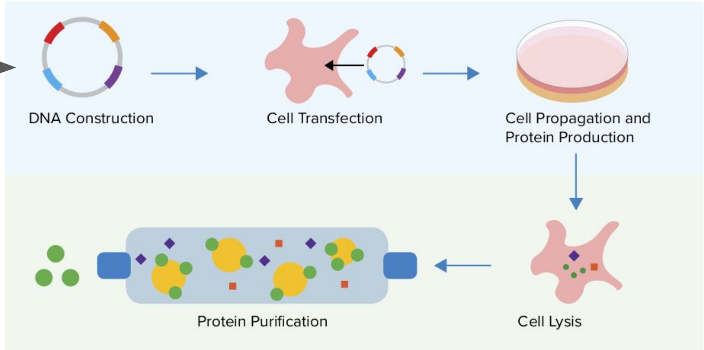
Cons :
Hallucinations

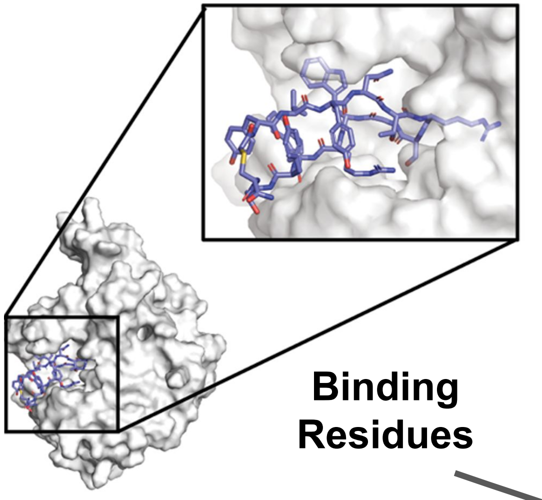
-2 TU | - 3 RU
+50 Novelty
+10 Confidence

Sequence Optimization

1 **Site-Saturation Mutagenesis**

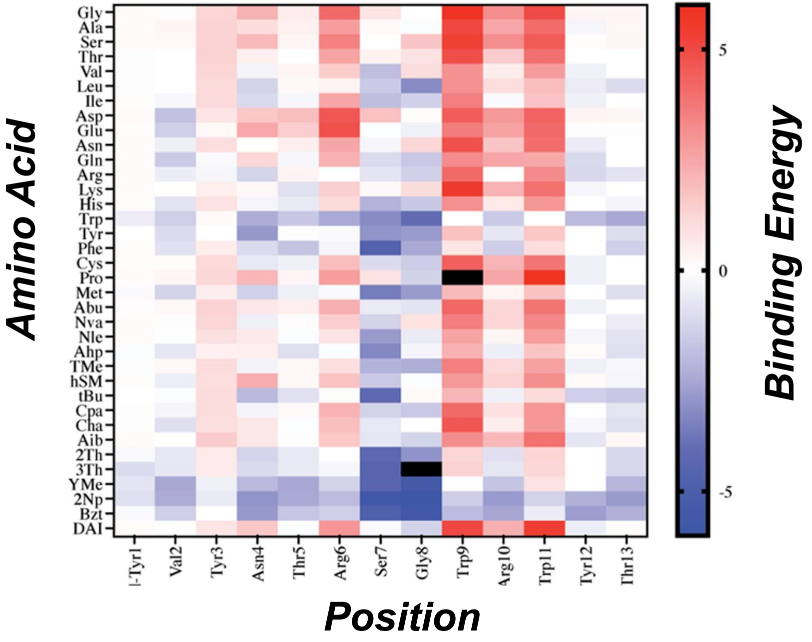
Mutating to all 20 amino acids





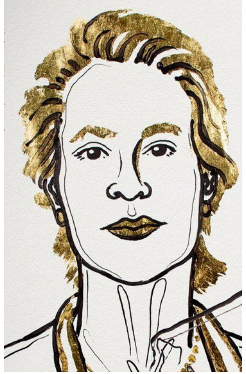
Binding Residues

Wild type	M	W	I	K	I	P	E	D
I5A	M	W	I	K	A	P	E	D
I5C	M	W	I	K	C	P	E	D
I5D	M	W	I	K	D	P	E	D
I5E	M	W	I	K	E	P	E	D
I5F	M	W	I	K	F	P	E	D
I5G	M	W	I	K	G	P	E	D
I5H	M	W	I	K	H	P	E	D
I5K	M	W	I	K	K	P	E	D
I5L	M	W	I	K	L	P	E	D
I5M	M	W	I	K	M	P	E	D
I5N	M	W	I	K	N	P	E	D
I5P	M	W	I	K	P	P	E	D
I5Q	M	W	I	K	Q	P	E	D
I5R	M	W	I	K	R	P	E	D
I5S	M	W	I	K	S	P	E	D
I5T	M	W	I	K	T	P	E	D
I5V	M	W	I	K	V	P	E	D
I5W	M	W	I	K	W	P	E	D
I5Y	M	W	I	K	Y	P	E	D



Sequence Optimization

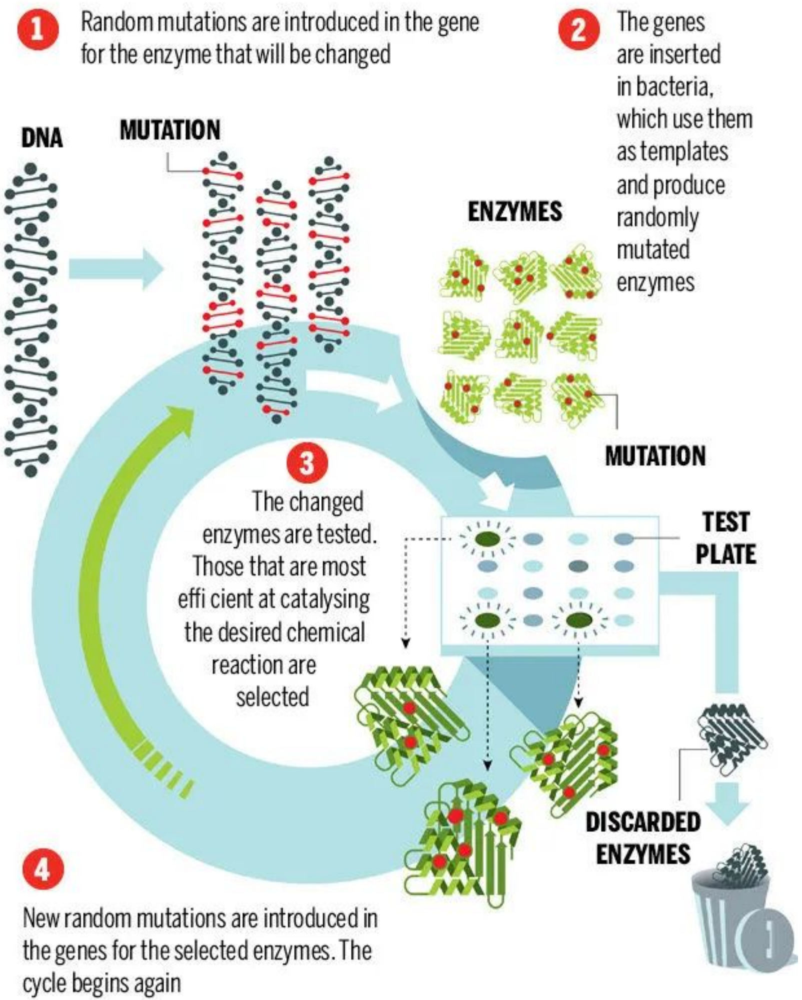
2 Directed Evolution



Frances H. Arnold

"for the directed evolution of enzymes"

THE NOBEL PRIZE IN CHEMISTRY 2018



Sequence Optimization

3 **AI Refinement**

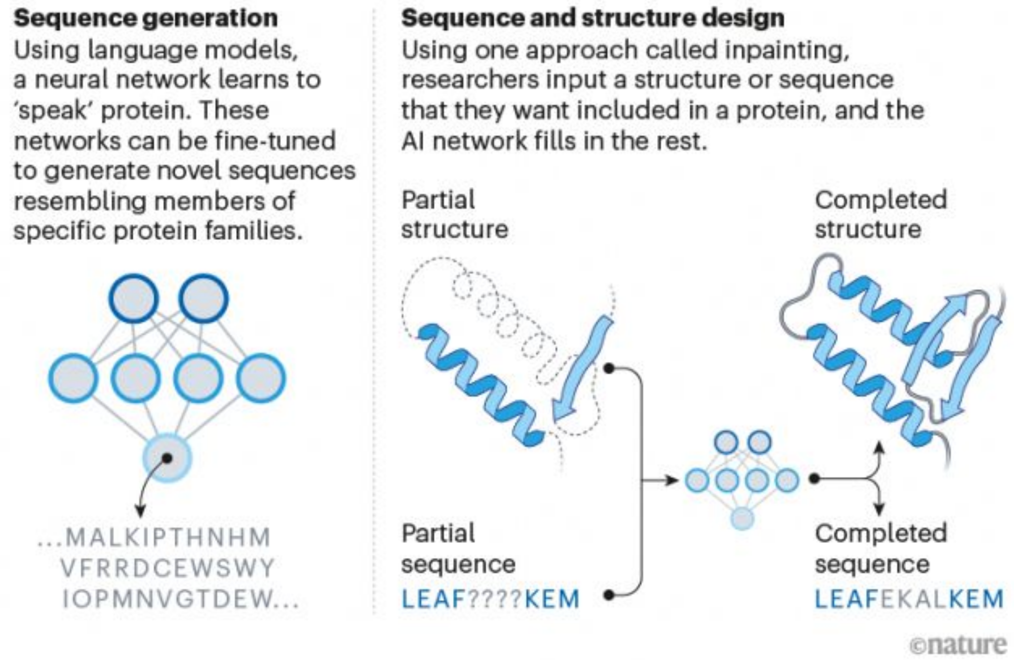
THE NOBEL PRIZE IN CHEMISTRY 2024

David Baker

"for computational protein design"

Rosetta Commons

ProteinMPNN



Sequence Optimization

1

Site-Saturation Mutagenesis

2

Directed Evolution

3

AI Refinement

Sequence Optimization

1

Site-Saturation Mutagenesis

Pros :
Exhaustive local mapping
Less complex protocol

Cons :
Blind to Epistasis
Scalability Limits

-2 TU | - 1 RU
+5 Novelty
+15 Confidence

2

Directed Evolution

Pros :
Massive combinations explored
Biological filter

Cons :
Selection bias & artifacts
Expression host discord

-1 TU | - 3 RU
+20 Novelty
+25 Confidence

3

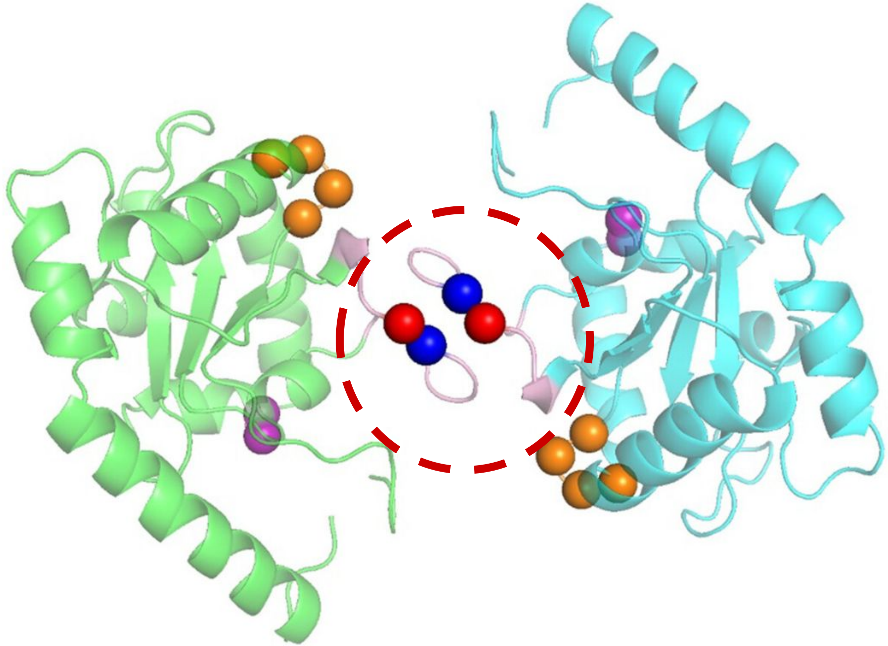
AI Refinement

Pros :
Global sequence optimization

Cons :
May prioritize stability over affinity

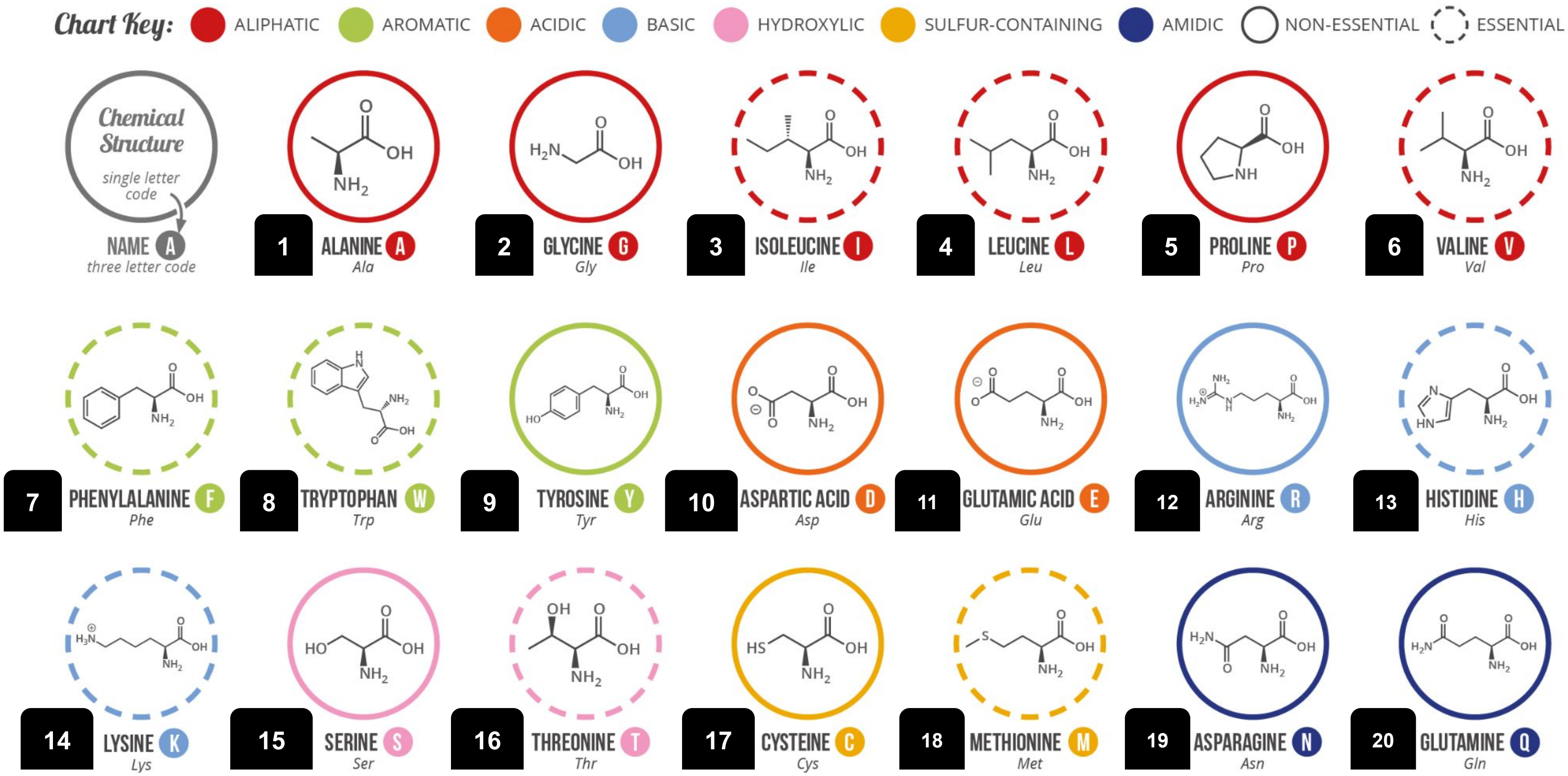
-1 TU | - 3 RU
+15 Novelty
+10 Confidence

Optimizing binding complementarity



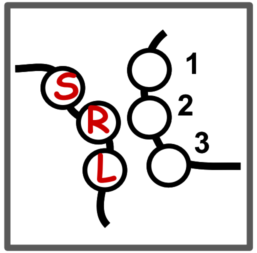
20 Standard Amino Acids

Choose the 3 amino acids at the binding site based on your birth date



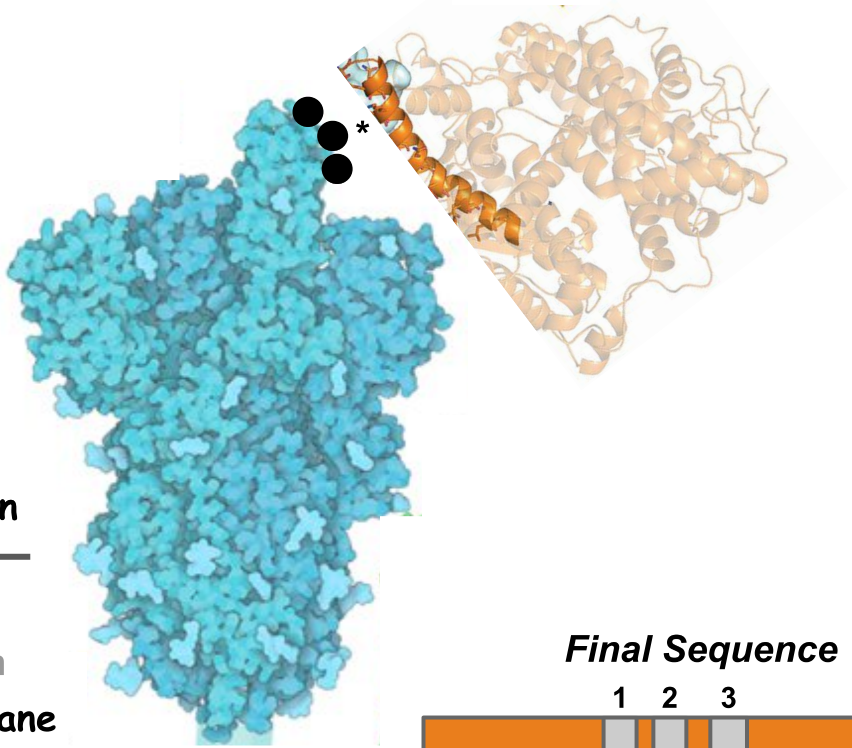
Goal : De Novo Design of Proteins to Neutralize SARS-CoV-2

Amino Acid Type	Amino Acid	1 Letter Code
Hydrophobic: Aliphatic	Glycine	G
	Alanine	A
	Valine	V
	Leucine	L
	Isoleucine	I
	Proline	P
	Methionine	M
Hydrophobic: Aromatic	Phenylalanine	F
	Tyrosine	Y
	Tryptophan	W
Hydrophilic: Polar Uncharged	Serine	S
	Threonine	T
	Cysteine	C
	Asparagine	N
	Glutamine	Q
Hydrophilic: Acidic	Aspartic Acid	D
	Glutamic Acid	E
Hydrophilic: Basic	Arginine	R
	Histidine	H
	Lysine	K



* Binding Interface

Scaffold



Target

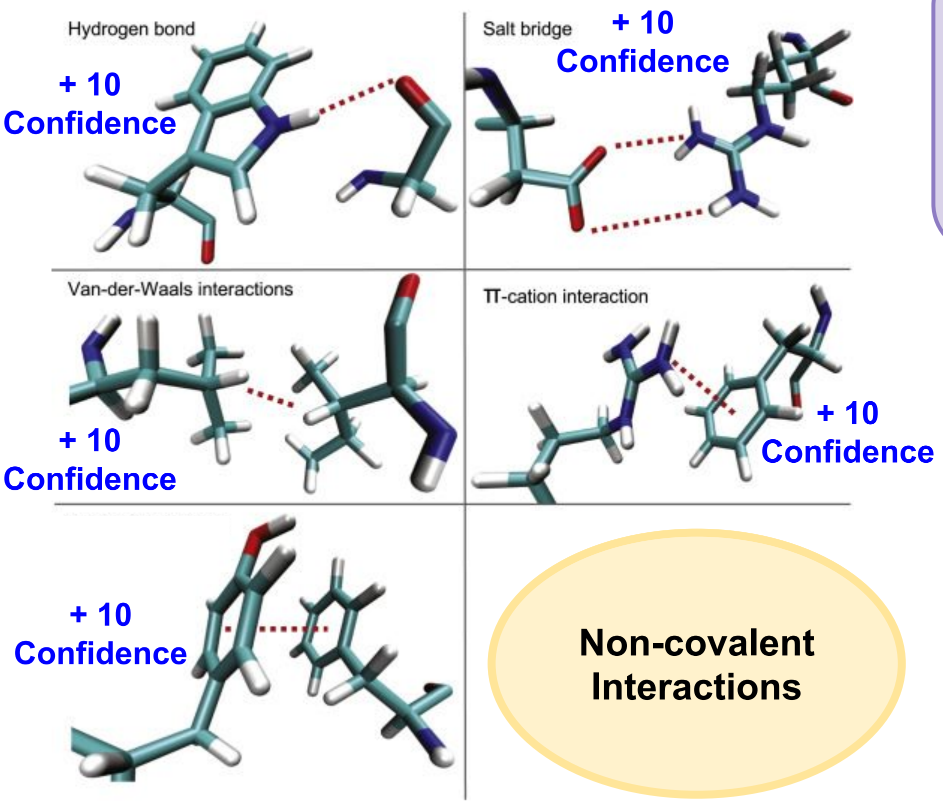
Spike Protein

Subcellular Localisation

Virion membrane

Final Sequence

Binding Site



Non-covalent Interactions

Unfavourable Electrostatics
Like charges interacting

- 10 Confidence

Aggregation Risk
>2 Hydrophobic residues in the binding site

- 10 Confidence

Rosetta Commons

PDBePISA

CAPS (Cys Ala Pro Ser)

DockScore
Scoring Server for docked poses

PPCheck

A webserver for identification of non-covalent interactions at the interface of a protein-protein complex.

Action Cards

A

Grant Secured

Your team successfully locked down a major collaborative institutional equipment grant, giving your team free access to high-end structural screening tools.

Gain +2 RU

B

Serendipitous Epistasis

A sequencing error reveals a spontaneous mutation at the interface that works cooperatively with an existing residue, dramatically locking down the binding loop.

Gain +10 Novelty

C

Host Cell Toxicity

The host bacteria find your newly designed miniprotein highly toxic. The cells lyse early, leaving you with nothing but a flask of sad, un-folded debris

Lose -30 Confidence

D

Side Project Detour

A co-author asks if you can “just quickly generate a better violin plot" for a figure on a side project, which turns into a grueling multi-hour battle against formatting and pixel margins.

Lose -1 TU

Validation

AlphaFold Structure

Check if the AI thinks the sequence will actually fold into your 3D shape

-1 TU | - 0 RU | + 15 Confidence

Energy Estimation

In silico calculations mapping the binding energy of the interface.

-1 TU | - 1 RU | + 20 Confidence

Stability and Folding

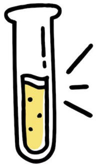
Circular dichroism spectroscopy to confirm it forms a stable and folded 3D structure

-2 TU | - 2 RU | + 25 Confidence

Binding Assay

Running an Y2H, ELISA or BLI to physically check if the protein bind to the target

-2 TU | - 3 RU | + 40 Confidence



In Vitro Neutralization

The ultimate test: mixing the protein with live cells to completely block infection

+ 60 Confidence

Goal : De Novo Design of Proteins to Neutralize SARS-CoV-2

De novo design of potent and resilient hACE2 decoys to neutralize SARS-CoV-2

THOMAS W. LINSKY, RENAN VERGARA, NURIA CODINA, JORGEN W. NELSON, MATTHEW J. WALKER, WEN SU, CHRISTOPHER O. BARNES, TIEN-YING HSIANG, KATHARINA ESSER-NOBIS, [...], AND DANIEL-ADRIANO SILVA +25 authors Authors Info & Affiliations

SCIENCE · 5 Nov 2020 · Vol 370, Issue 6521 · pp. 1208-1214 · DOI: 10.1126/science.abe0075

SARS CoV-2 Nucleoprotein Enhances the Infectivity of Lentiviral Spike Particles

Tarun Mishra[1†], M. Sreepadmanabh[1†], Pavitra Ramdas[1], Amit Kumar Sahu[1], Atul Kumar[2] and Ajit Chande[1]*

De novo design of picomolar SARS-CoV-2 miniprotein inhibitors

LONGXING CAO, INNA GORESHNIK, BRIAN COVENTRY, JAMES BRETT CASE, LAUREN MILLER, LISA KOZODOY, RITA E. CHEN, LAUREN CARTER, ALEXANDRA C. WALLS, [...], AND DAVID BAKER +5 authors Authors Info & Affiliations

SCIENCE · 9 Sep 2020 · Vol 370, Issue 6515 · pp. 426-431 · DOI: 10.1126/science.abd9909

De novo generation of SARS-CoV-2 antibody CDRH3 with a pre-trained generative large language model

Haohuai He, Bing He, Lei Guan, Yu Zhao, Feng Jiang, Guanxing Chen, Qingge Zhu, Calvin Yu-Chian Chen, Ting Li & Jianhua Yao

Nature Communications **15**, Article number: 6867 (2024) | Cite this article

Summary

Protein design =
function --> structure --> sequence

- **Proteins are dynamic**, but so is the process of designing them :

- **Research is dynamic** - you are always switching between roles, especially the 4 each team were given

- Modern protein research is about **discovering as well as creating** new sequences, structures and functions; but it is also about -
  - Managing real life expectations and limitations
  - Community
    - Designing for benefit
    - Sharing resources
    - Respectful peer review

DIY

Identify a 2026 global problem (e.g., oil spill cleanup, sensing gluten in food, or neutralising a new pathogen) that you can tackle by designing a new protein.

Define the protein's mission and propose a design plan

References

(for images used)

- **https://www.ibiology.org/sessions/session-3-protein-design/**
- https://bioboot.github.io/bimm143_S24/class-material/class11_alphafold.html
- https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2017.01557/full
- https://www.fpm.org.uk/blog/covid-19-sars-cov-2-pandemic/
- **https://pdb101.rcsb.org/learn/flyers-posters-and-calendars/flyer/sars-cov-2-genome-and-proteins**
- https://pdb101.rcsb.org/motm/21
- https://www.sciencedirect.com/science/article/pii/S0959440X26000060
- https://www.bakerlab.org/2023/03/30/rf-diffusion-now-free-and-open-source/
- https://www.rcsb.org/news/5bb4c7d49a2ba162ac026c38
- https://www.nobelprize.org/prizes/chemistry/2018/press-release/
- https://tacc.utexas.edu/news/latest-news/2024/10/10/frontera-supported-protein-design-nets-chemistry-nobel-prize/
- https://rosettacommons.org/
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That’s all folks !